

Machine Learning

CSE343/CSE543/ECE363/ECE563

Lecture 10 | Take your own notes during lectures

Vinayak Abrol <abrol@iiitd.ac.in>



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY **DELHI**



SVM: The Optimal Margin Classifier

Assumption: training set that is linearly separable.

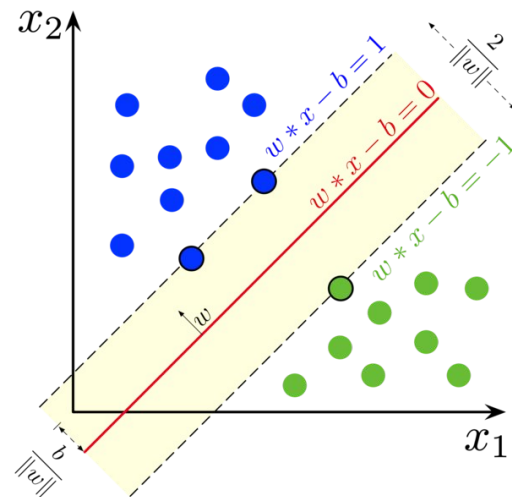
$$\begin{aligned} \max_{\gamma, w, b} \quad & \gamma \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq \gamma, \quad i = 1, \dots, m \\ & \|w\| = 1. \end{aligned}$$

Solve this and we are done!

But $\|w\| = 1$ is a non-convex constraint. Why?

$$\begin{aligned} \max_{\gamma, w, b} \quad & \frac{\hat{\gamma}}{\|w\|} & \gamma = \hat{\gamma} / \|w\| \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq \hat{\gamma}, \quad i = 1, \dots, m \end{aligned}$$

Recall the scaling property of functional margin!



$$\hat{\gamma} / \|w\| = 1 / \|w\|$$

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1, \quad i = 1, \dots, m \end{aligned}$$

We use Quadratic Programming (QP) solvers

Lagrange Multiplier

Solving constrained optimization problems

$$\begin{array}{ll}\min_w & f(w) \\ \text{s.t.} & h_i(w) = 0, \quad i = 1, \dots, l\end{array}$$

Solve this using [Method of Lagrange Multipliers](#)

$$\mathcal{L}(w, \beta) = f(w) + \sum_{i=1}^l \beta_i h_i(w)$$

Lagrange Multiplier

Solving constrained optimization problems

$$\begin{aligned} \min_w \quad & f(w) \\ \text{s.t.} \quad & h_i(w) = 0, \quad i = 1, \dots, l \end{aligned}$$

Solve this using **Method of Lagrange Multipliers**

$$\mathcal{L}(w, \beta) = f(w) + \sum_{i=1}^l \beta_i h_i(w)$$

β are called the Lagrange multipliers

$\mathcal{L}(w, \beta)$ is called Lagrange function

$$\frac{\partial \mathcal{L}}{\partial w_i} = 0; \quad \frac{\partial \mathcal{L}}{\partial \beta_i} = 0$$

Take Gradients and solve for (w, β)

Lagrange Multiplier

Solving constrained optimization problems

$$\begin{aligned} \min_w \quad & f(w) \\ \text{s.t.} \quad & h_i(w) = 0, \quad i = 1, \dots, l \end{aligned}$$

Solve this using **Method of Lagrange Multipliers**

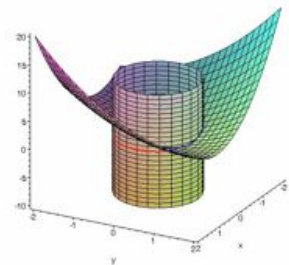
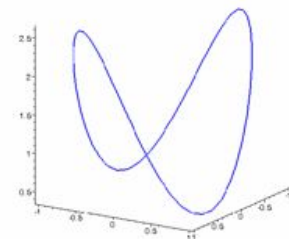
$$\mathcal{L}(w, \beta) = f(w) + \sum_{i=1}^l \beta_i h_i(w)$$

β are called the Lagrange multipliers

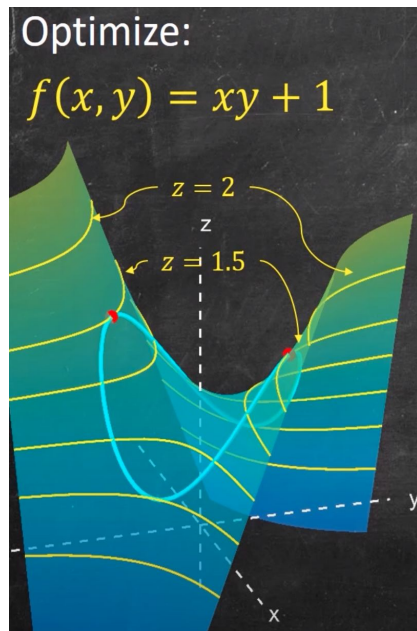
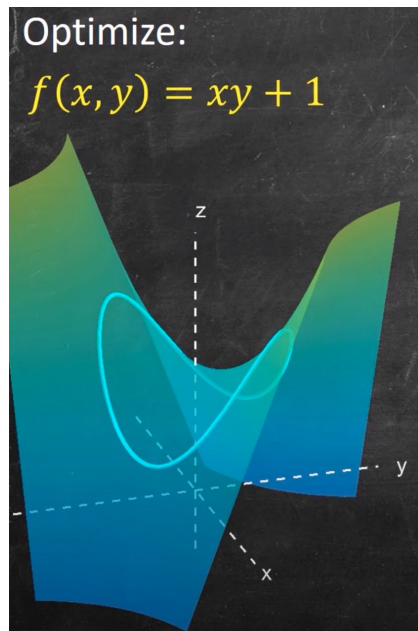
$\mathcal{L}(w, \beta)$ is called Lagrange function

Take Gradients and solve for (w, β)

$$\frac{\partial \mathcal{L}}{\partial w_i} = 0; \quad \frac{\partial \mathcal{L}}{\partial \beta_i} = 0$$



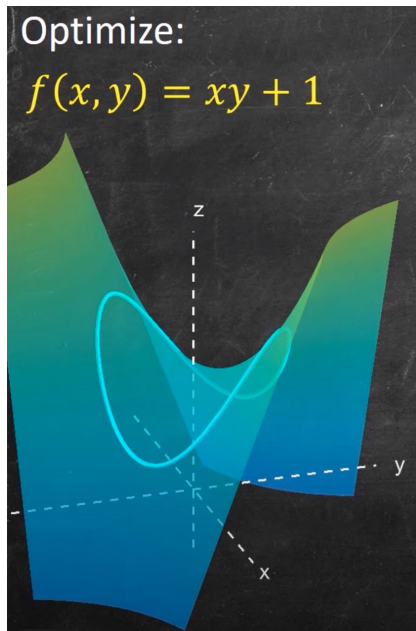
Lagrange Method: Visualization



Lagrange Method: Visualization

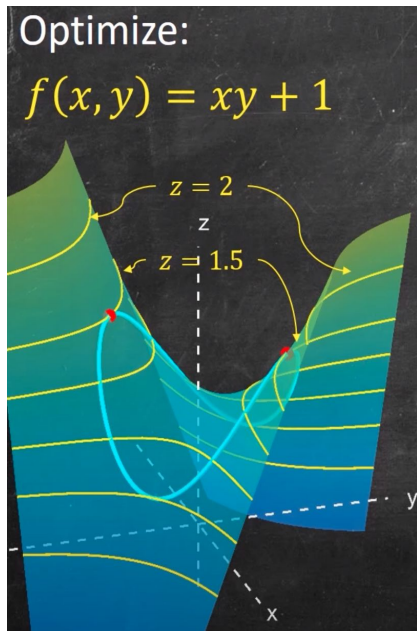
Optimize:

$$f(x, y) = xy + 1$$



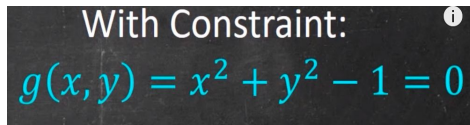
Optimize:

$$f(x, y) = xy + 1$$



With Constraint:

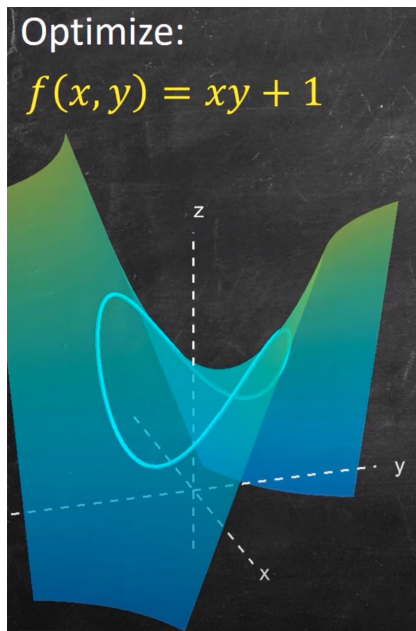
$$g(x, y) = x^2 + y^2 - 1 = 0$$



Lagrange Method: Visualization

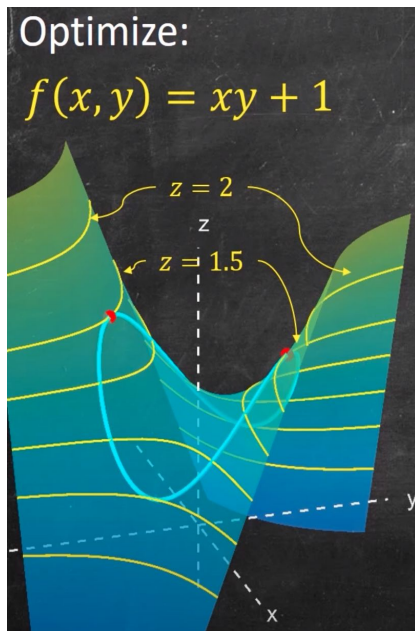
Optimize:

$$f(x, y) = xy + 1$$



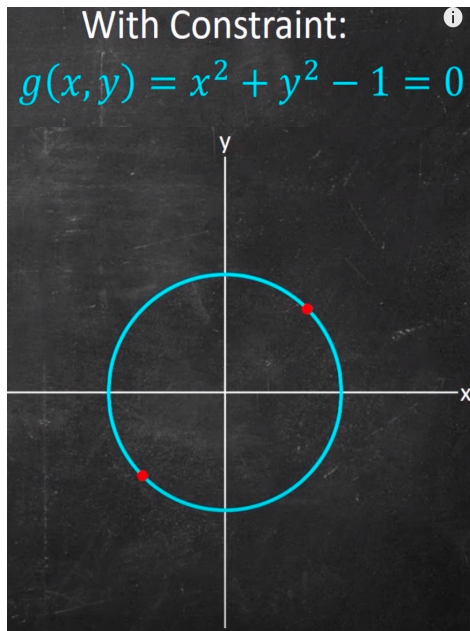
Optimize:

$$f(x, y) = xy + 1$$



With Constraint:

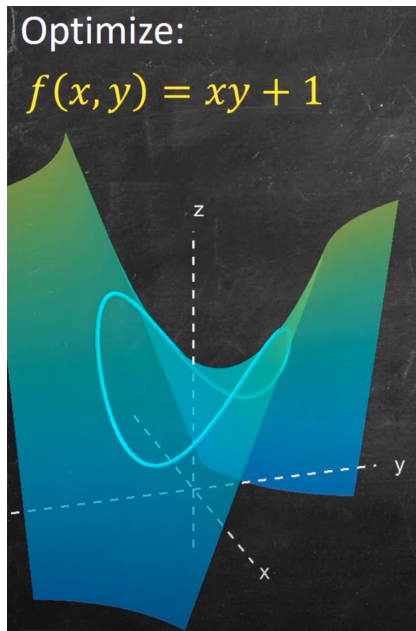
$$g(x, y) = x^2 + y^2 - 1 = 0$$



Lagrange Method: Visualization

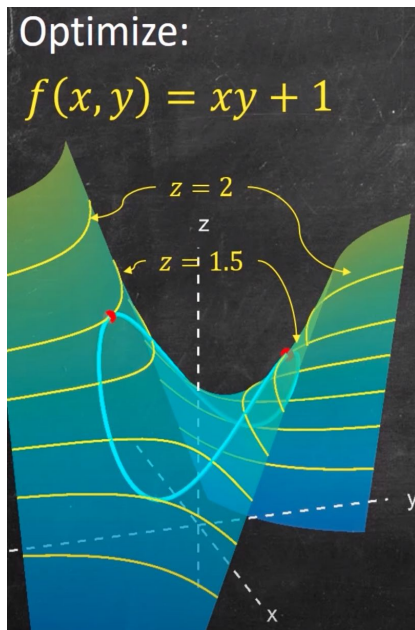
Optimize:

$$f(x, y) = xy + 1$$



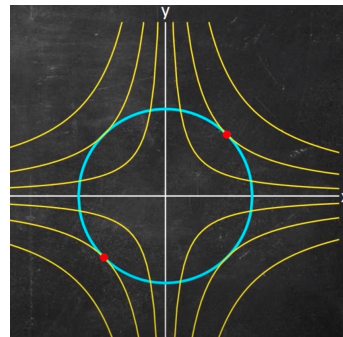
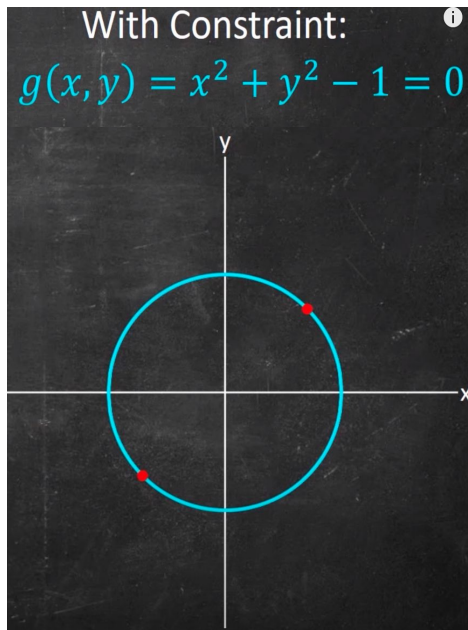
Optimize:

$$f(x, y) = xy + 1$$



With Constraint:

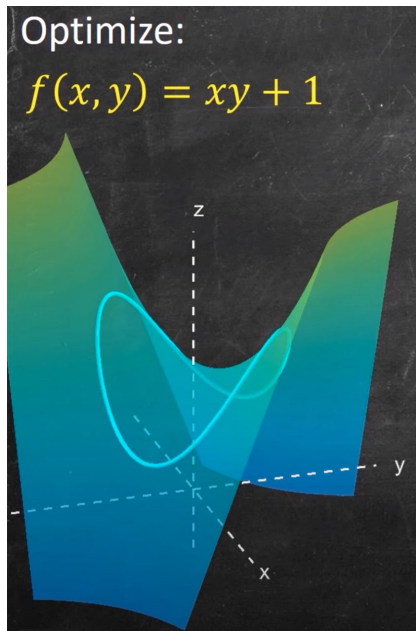
$$g(x, y) = x^2 + y^2 - 1 = 0$$



Lagrange Method: Visualization

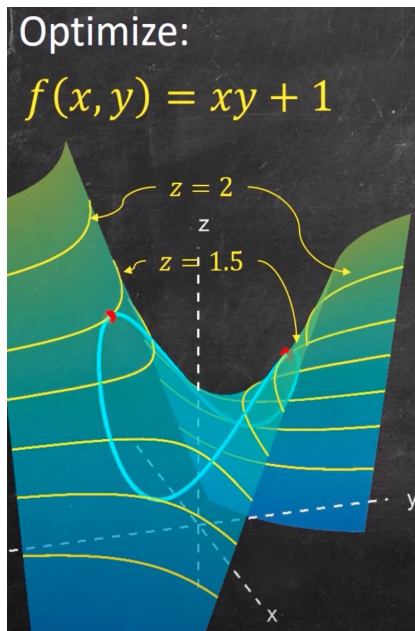
Optimize:

$$f(x, y) = xy + 1$$



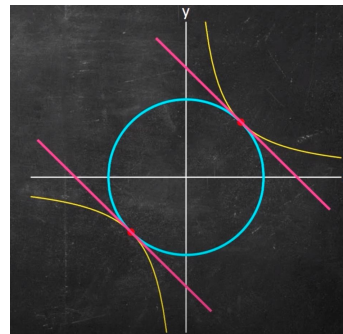
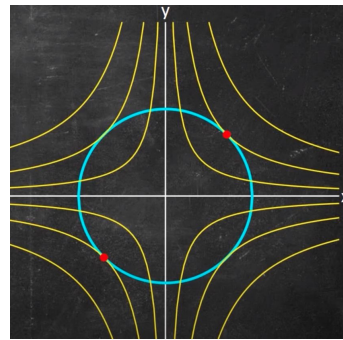
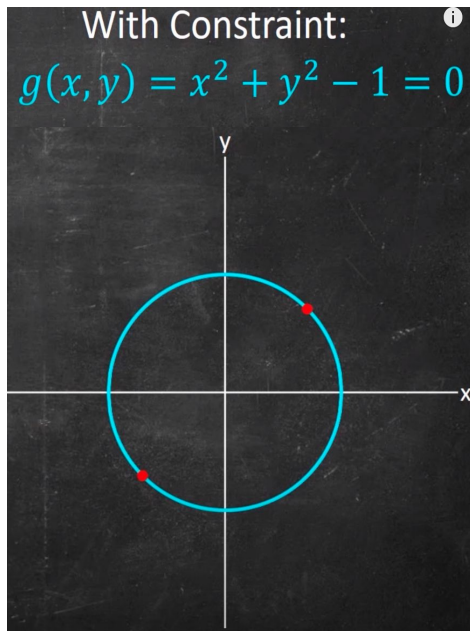
Optimize:

$$f(x, y) = xy + 1$$

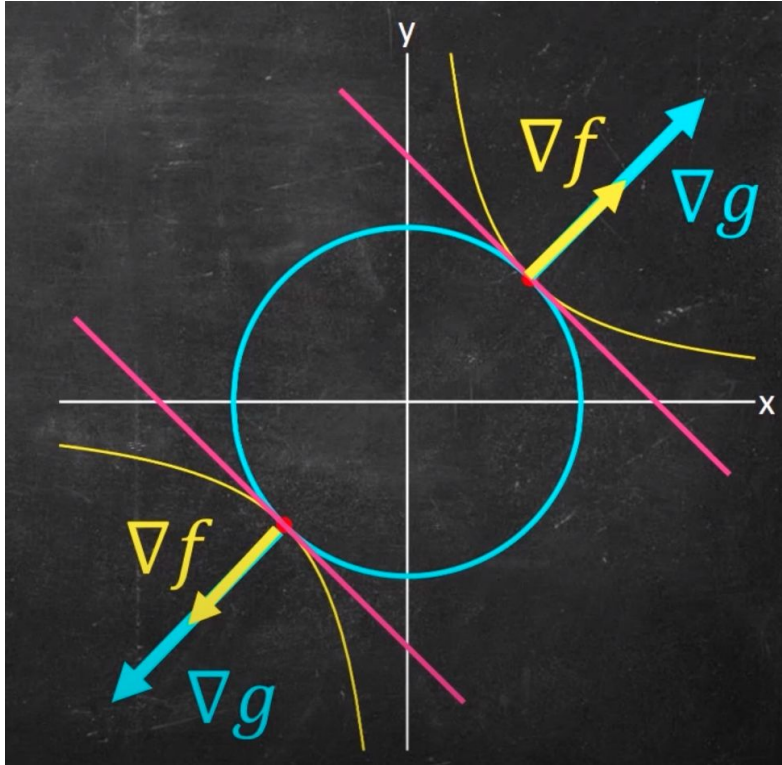


With Constraint:

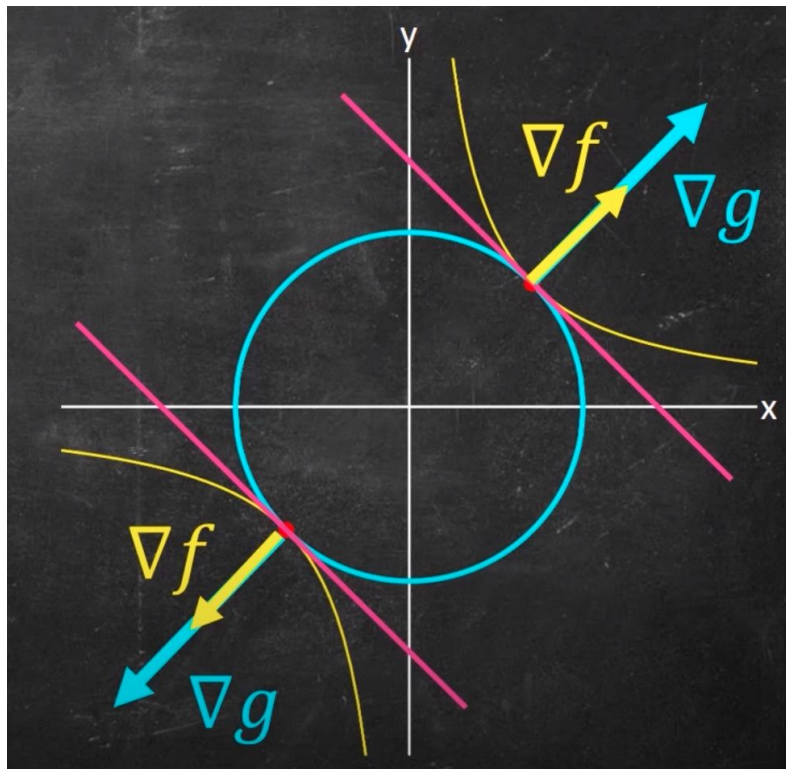
$$g(x, y) = x^2 + y^2 - 1 = 0$$



Lagrange Method: Visualization



Lagrange Method: Visualization



Lagrange Multipliers:
Simultaneously solve

$$\nabla f = \lambda \nabla g$$
$$g = 0$$

Primal and Dual Problem

$$\begin{array}{ll}\min_w & f(w) \\ \text{s.t.} & g_i(w) \leq 0, \quad i = 1, \dots, k \\ & h_i(w) = 0, \quad i = 1, \dots, l.\end{array}$$

$$\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w).$$

Primal and Dual Problem

$$\begin{array}{ll}\min_w & f(w) \\ \text{s.t.} & g_i(w) \leq 0, \quad i = 1, \dots, k \\ & h_i(w) = 0, \quad i = 1, \dots, l.\end{array}$$

$$\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w).$$

$$\theta_{\mathcal{P}}(w) = \max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta)$$

Primal

$$\min_w \theta_{\mathcal{P}}(w) = \min_w \max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta)$$

$$\theta_{\mathcal{P}}(w) = \begin{cases} f(w) & \text{if } w \text{ satisfies primal constraints} \\ \infty & \text{otherwise.} \end{cases}$$

Primal and Dual Problem

$$\begin{array}{ll}\min_w & f(w) \\ \text{s.t.} & g_i(w) \leq 0, \quad i = 1, \dots, k \\ & h_i(w) = 0, \quad i = 1, \dots, l.\end{array}$$

$$\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w).$$

Primal

$$\min_w \theta_{\mathcal{P}}(w) = \min_w \max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta)$$

$$\theta_{\mathcal{P}}(w) = \begin{cases} f(w) & \text{if } w \text{ satisfies primal constraints} \\ \infty & \text{otherwise.} \end{cases}$$

Dual

$$\max_{\alpha, \beta: \alpha_i \geq 0} \theta_{\mathcal{D}}(\alpha, \beta) = \max_{\alpha, \beta: \alpha_i \geq 0} \min_w \mathcal{L}(w, \alpha, \beta)$$

$$d^* = \max_{\alpha, \beta: \alpha_i \geq 0} \min_w \mathcal{L}(w, \alpha, \beta) \leq \min_w \max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta) = p^*$$

Primal and Dual Problem

$$\begin{aligned} \min_w \quad & f(w) \\ \text{s.t.} \quad & g_i(w) \leq 0, \quad i = 1, \dots, k \\ & h_i(w) = 0, \quad i = 1, \dots, l. \end{aligned}$$

Primal

$$\min_w \theta_{\mathcal{P}}(w) = \min_w \max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta)$$

$$\theta_{\mathcal{P}}(w) = \begin{cases} f(w) & \text{if } w \text{ satisfies primal constraints} \\ \infty & \text{otherwise.} \end{cases}$$

$$\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w).$$

Dual

$$\begin{aligned} \frac{\partial}{\partial w_i} \mathcal{L}(w^*, \alpha^*, \beta^*) &= 0, \quad i = 1, \dots, n \\ \frac{\partial}{\partial \beta_i} \mathcal{L}(w^*, \alpha^*, \beta^*) &= 0, \quad i = 1, \dots, l \\ \alpha_i^* g_i(w^*) &= 0, \quad i = 1, \dots, k \\ g_i(w^*) &\leq 0, \quad i = 1, \dots, k \\ \alpha^* &\geq 0, \quad i = 1, \dots, k \end{aligned}$$

$d^* = p^*$ when (w, α, β) satisfy
Karush-Kun-Tucker Conditions

$$d^* = \max_{\alpha, \beta: \alpha_i \geq 0} \min_w \mathcal{L}(w, \alpha, \beta) \leq \min_w \max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta) = p^*$$

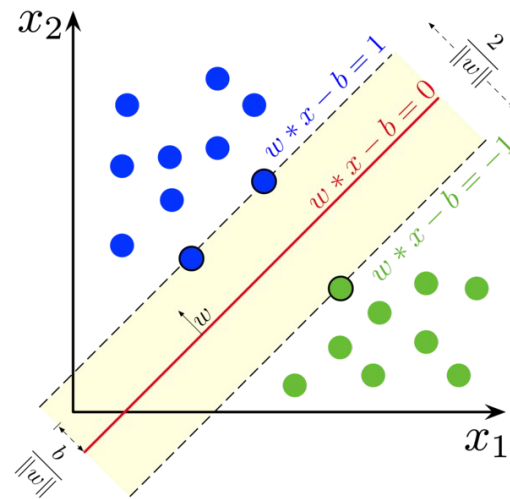
SVM: The Optimal Margin Classifier

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1, \quad i = 1, \dots, m \end{aligned}$$

Consider the 3 points (2 Blue) and (1 Green)
Only these 3 points satisfy the constraints!

These are called as **Support Vectors**

Let's find them using duality

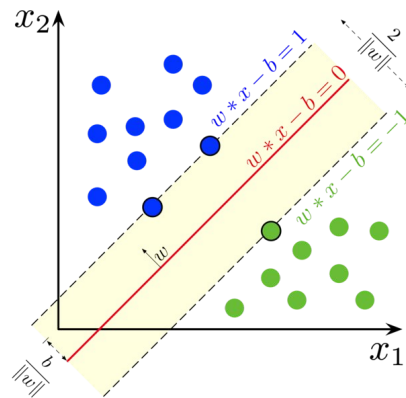


SVM: The Optimal Margin Classifier

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1, \quad i = 1, \dots, m \end{aligned}$$

$$g_i(w) = -y^{(i)}(w^T x^{(i)} + b) + 1 \leq 0$$

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^m \alpha_i [y^{(i)}(w^T x^{(i)} + b) - 1]$$

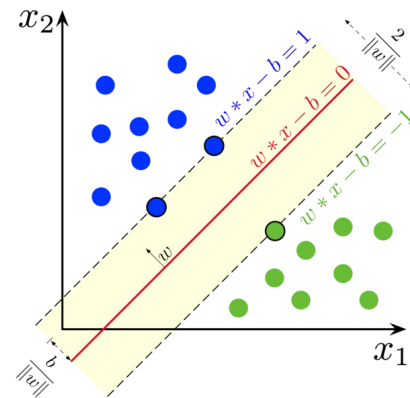


SVM: The Optimal Margin Classifier

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1, \quad i = 1, \dots, m \end{aligned}$$

$$g_i(w) = -y^{(i)}(w^T x^{(i)} + b) + 1 \leq 0$$

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^m \alpha_i [y^{(i)}(w^T x^{(i)} + b) - 1]$$



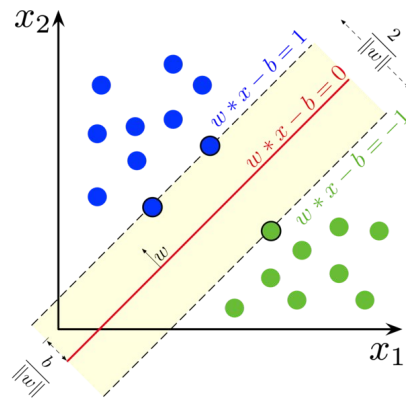
For dual form of the problem, we need to first minimize $L(w, b, \alpha)$ with respect to w and b (for fixed α)

SVM: The Optimal Margin Classifier

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1, \quad i = 1, \dots, m \end{aligned}$$

$$g_i(w) = -y^{(i)}(w^T x^{(i)} + b) + 1 \leq 0$$

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^m \alpha_i [y^{(i)}(w^T x^{(i)} + b) - 1]$$



For dual form of the problem, we need to first minimize $L(w, b, \alpha)$ with respect to w and b (for fixed α)
Let's take gradients

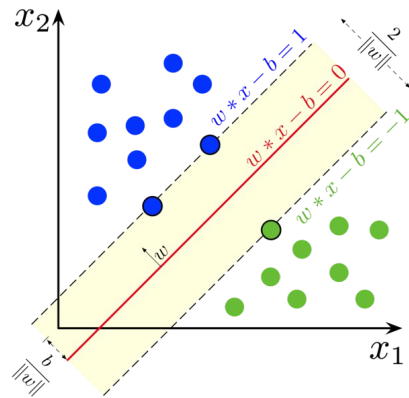
$$\nabla_w \mathcal{L}(w, b, \alpha) = w - \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} = 0$$

$$w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}.$$

$$\frac{\partial}{\partial b} \mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i y^{(i)} = 0.$$

SVM: The Optimal Margin Classifier

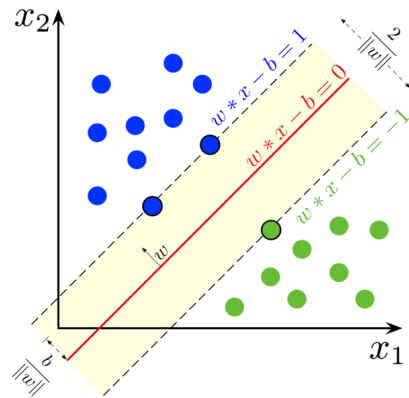
$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)} - b \sum_{i=1}^m \alpha_i y^{(i)}$$



SVM: The Optimal Margin Classifier

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)} - b \sum_{i=1}^m \alpha_i y^{(i)}$$

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)}.$$



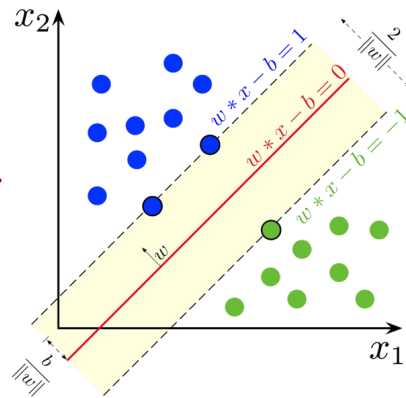
SVM: The Optimal Margin Classifier

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)} - b \sum_{i=1}^m \alpha_i y^{(i)}$$

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)}.$$

KKT conditions are satisfied.
Verify!

$$\begin{aligned} \max_{\alpha} \quad & W(\alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle. \\ \text{s.t.} \quad & \alpha_i \geq 0, \quad i = 1, \dots, m \\ & \sum_{i=1}^m \alpha_i y^{(i)} = 0, \end{aligned}$$



SVM: The Optimal Margin Classifier

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)} - b \sum_{i=1}^m \alpha_i y^{(i)}$$

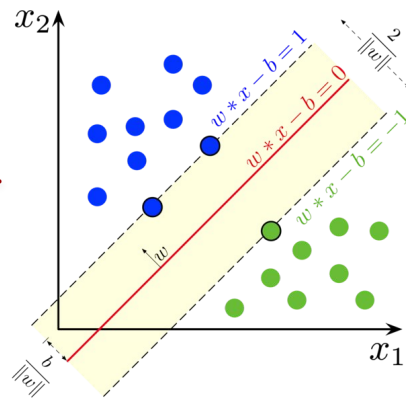
$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)}.$$

KKT conditions are satisfied.
Verify!

$$\begin{aligned} \max_{\alpha} \quad & W(\alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle. \\ \text{s.t.} \quad & \alpha_i \geq 0, \quad i = 1, \dots, m \\ & \sum_{i=1}^m \alpha_i y^{(i)} = 0, \end{aligned}$$

Consider a test example with $y=1$. Here we use the value of w in terms of α

$$\begin{aligned} w^T x + b &= \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T x + b \\ &= \sum_{i=1}^m \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b. \end{aligned}$$



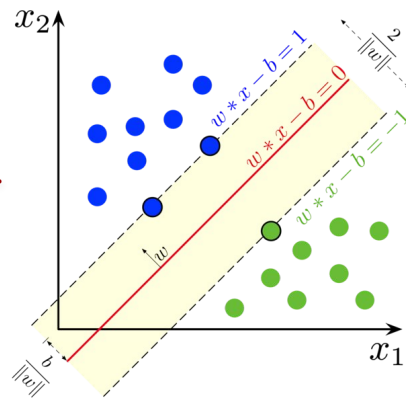
SVM: The Optimal Margin Classifier

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)} - b \sum_{i=1}^m \alpha_i y^{(i)}$$

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)}.$$

KKT conditions are satisfied.
Verify!

$$\begin{aligned} \max_{\alpha} \quad & W(\alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle. \\ \text{s.t.} \quad & \alpha_i \geq 0, \quad i = 1, \dots, m \\ & \sum_{i=1}^m \alpha_i y^{(i)} = 0, \end{aligned}$$



Consider a test example with $y=1$. Here we use the value of w in terms of α

$$\begin{aligned} w^T x + b &= \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T x + b \\ &= \sum_{i=1}^m \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b. \end{aligned}$$

Making a prediction requires the inner-product $\langle x^{(i)}, x \rangle$
Multiplied with $\alpha_i > 0$ i.e., support vectors.

SVM: Maths Looks Heavy?

To be continued.....

